

MMI 714 - Generative Models for Multimedia

Project Proposal

Brain As a Generative Model : Exploration of the Intuitive Physics through Latent Space Disentanglement

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Abstract :

The primary objective of this project is to explore *intuitive physics*, which encompasses an understanding of fundamental physical principles, such as object permanence (Lake et al., 2017). Intuitive physics research has been approached from diverse perspectives and domains, including cognitive science (Lake et al., 2017; Battaglia et al., 2013), developmental psychology (Piloto et al., 2022), and deep learning engineering (Lerer et al., 2016). Prominent contributions include Battaglia et al.'s Intuitive Physics Engine (IPE), which leverages Bayesian cognitive modeling to answer questions like "Will it fall?" and "In which direction?" (Battaglia et al., 2013), and also Piloto et al.'s study highlighting the gap in intuitive physics capabilities between humans and AI systems (Piloto et al., 2022).

Despite the diversity of approaches, existing studies primarily rely on traditional deep learning methods, such as Convolutional Neural Networks (CNNs) for stability prediction (Lerer et al., 2016), without leveraging latent space disentanglement for intuitive physics. This project proposes training a *Variational Autoencoder (VAE)* to disentangle the latent space, enabling controlled exploration of physical dynamics in synthetic block-stacking scenarios. The aim is to model intuitive physics by incorporating spatial and temporal dynamics, disentangling latent factors for key attributes such as velocity, position, and time.

Implementation of the Project:

1. Dataset Generation :

Synthetic data will be created using the Pygame and Pymunk libraries, capturing 20-second video sequences of falling block towers alongside metadata (object coordinates, velocities, and time). While existing datasets, such as ShapeStack, are designed for intuitive physics, they lack temporal information, consisting solely of static images. This limitation necessitates custom data generation to incorporate temporal dynamics essential for disentangling time-related latent factors.

The generated dataset will enable precise control over the temporal axis of the latent space, facilitating the disentanglement of dependent components such as velocity and position over time.

2. Model Architecture :

To achieve a disentangled latent space, the following architecture will be employed:

1. **Variational Autoencoder (VAE):** A VAE will be trained to encode video sequences into a latent space, with explicit disentanglement of spatial (position, shape) and temporal (velocity, time progression) factors.
2. **Hybrid Encoder:**
 - **Convolutional Neural Networks (CNNs)** will process spatial features within individual video frames.
 - **Recurrent Neural Networks (RNNs)**, such as LSTMs, will model temporal dependencies across frames, processing velocity and coordinate data in conjunction with frame encodings.
3. **Multi-modal Input Handling:** Metadata (e.g., velocity, coordinates) will be integrated into the encoder alongside frame data to enhance the semantic structure of the latent space

3. Loss Functions and Variational Approaches :

To encourage latent space disentanglement and ensure meaningful representations, the following loss functions and techniques will be used:

1. **Beta-VAE:** Introduce a hyperparameter ($\beta > 1$) to encourage disentanglement in the KL divergence term, promoting independent latent dimensions.
2. **Factor-VAE:** Includes a regularization term to penalize correlations among latent dimensions, ensuring factorized representations.
3. **Conditional VAE:** Encode metadata (like time, velocity) alongside frames to make the latent space semantically meaningful.

4. Validation Metrics :

Since intuitive physics lacks a formal mathematical definition, model validation will involve comparison with human performance on equivalent tasks. Participants will be presented with data from latent space, and their responses will be compared with the latent traversals' outputs. Similar validation approaches (comparing with humans) have been employed in related studies (Battaglia et al., 2013; Lerer et al., 2016), offering a robust benchmark for evaluating intuitive physics models. Additionally, qualitative analyses of the latent space (e.g., latent traversals) will confirm whether individual dimensions correspond to interpretable physical attributes, such as object trajectories or temporal dynamics.

Conclusion

This project aims to bridge a significant gap in the study of intuitive physics by employing latent space disentanglement to model physical dynamics in a controllable and interpretable manner. By integrating spatial and temporal data into a unified latent representation, this work seeks to advance the understanding of how AI systems can emulate human-like intuitive physics. The proposed framework has implications for cognitive science, AI research, and applications requiring physical reasoning, such as robotics and virtual environments.